Observability Metric Examples—Enterprise GenAI Platform

Version 2026.1 | Azure Monitor, Application Insights, Azure AI Foundry | LC 3 Project Module

Overview

This document provides production-ready metric definitions, telemetry schemas, and dashboard specifications for monitoring a multi-application GenAI platform on Azure. All metrics align with Microsoft Azure Monitor 2026, OpenTelemetry semantic conventions, and the Azure AI Foundry observability model. Metrics are categorized by signal type: infrastructure, application performance, LLM-specific quality, cost, and security/compliance.

1. Distributed Tracing Architecture

Trace context must propagate across all components using W3C Trace Context headers. Azure Application Insights (native OpenTelemetry support) is the primary trace backend.

Trace Span Hierarchy Template:

```
Trace: askretail-conv-abc123
├── Span: gateway-request (Azure API Management) [duration: 12ms]
│   ├── Span: auth-validation (Entra ID) [duration: 8ms]
│   ├── Span: rate-limit-check (APIM cache) [duration: 2ms]
│   └── Span: backend-route (APIM policy) [duration: 2ms]
├── Span: orchestrator-request (AKS pod: askretail-v2.3.1) [duration: 45ms]
│   ├── Span: prompt-load (Azure App Config) [duration: 5ms]
│   ├── Span: retrieval-query (Azure AI Search) [duration: 35ms]
│   │   ├── Span: vector-search [duration: 22ms]
│   │   └── Span: semantic-rerank [duration: 13ms]
│   ├── Span: model-inference (Azure OpenAI) [duration: 680ms]
│   │   ├── Span: tokenization [duration: 8ms]
│   │   ├── Span: inference-core [duration: 640ms]
│   │   └── Span: post-processing [duration: 32ms]
│   └── Span: response-formatting [duration: 12ms]
└── Span: logging-audit (Azure Blob / Log Analytics) [duration: 8ms]
```

1.1 Required Trace Attributes (OpenTelemetry)

Attribute Key	Type	Example Value	Purpose
`ai.application.name`	string	askretail	Service name for app identification
`ai.application.version`	string	2.3.1	Deployed version for regression correlation
`ai.conversation.id`	string	conv-abc123	End-user conversation correlation
`ai.model.name`	string	gpt-4o-2026-01-15	Exact model version (no "latest")
`ai.model.provider`	string	azure_openai	Model hosting provider
`ai.prompt.tokens`	int	142	Input token count
`ai.completion.tokens`	int	89	Output token count
`ai.total.tokens`	int	231	Total token consumption
`ai.groundedness.score`	float	0.94	Groundedness metric (0–1)
`ai.hallucination.flag`	boolean	false	Binary hallucination detection
`ai.retrieval.source`	string	product-catalog-index-v3	Index or data source name
`ai.user.tier`	string	premium	User segmentation for SLA tracking
`http.response.status_code`	int	200	Standard HTTP status
`cloud.region`	string	eastus	Azure region for latency analysis

2. LLM-Specific Quality Metrics

These metrics require custom instrumentation via the Azure AI Evaluation SDK, Prompt Flow runtime exporters, or custom telemetry processors. All scores are normalized to 0–1 unless otherwise specified.

2.1 Groundedness

Definition: Measures whether the model response is factually grounded in the provided retrieval context (source documents).

- Computation Method: Azure AI Evaluation SDK `groundedness` evaluator. Uses NLI (Natural Language Inference) model to classify each claim in the response as supported, contradicted, or neutral relative to retrieved passages. Score = supported claims / total claims.
- Data Sources: Retrieved passages (Azure AI Search results), model response, conversation ID.
- Thresholds: AskRetail ≥ 0.92; HRHelper ≥ 0.90; StockSense N/A (no retrieval for forecasting).
- Alert Rule: Groundedness < threshold for > 5% of requests in 10-minute window → P2 alert to App Owner + Platform Ops.

2.2 Hallucination Detection

Definition: Binary or rate-based detection of fabricated information not present in retrieval context or model training data.

- Computation Method: Composite approach: (1) Azure AI Content Safety groundedness + self-consistency check (sample 3 temperature variants, measure semantic similarity); (2) Custom fine-tuned classifier for domain-specific hallucination patterns (product specs, HR policies).
- Data Sources: Model response, retrieval context, historical ground-truth dataset (curated QandA pairs).
- Thresholds: AskRetail < 1.5% of responses; HRHelper < 2.0%; StockSense N/A.
- Alert Rule: Hallucination rate > 3% in any 30-minute window → P1 alert + auto-canary halt for Tier 1 apps.

2.3 Semantic Drift

Definition: Detects when the distribution of model inputs or outputs shifts significantly from the training/evaluation baseline, indicating potential data drift or model degradation.

- Computation Method: Embed inputs and outputs using text-embedding-3-large. Compute cosine similarity distributions against baseline embeddings (Kolmogorov-Smirnov test or Wasserstein distance). Track per-topic cluster drift.
- Data Sources: 7-day rolling sample of user queries and model responses (anonymized). Baseline: evaluation dataset embeddings.
- Thresholds: KS statistic > 0.15 or Wasserstein distance > 0.3 → drift detected.
- Alert Rule: Drift detected for > 2 consecutive hours → P2 alert + trigger continuous evaluation replay.

2.4 Response Completeness and Relevance

Definition: Measures whether the response fully addresses the user intent and is relevant to the query.

- Computation Method: Azure AI Evaluation SDK `relevance` and `coherence` evaluators. Relevance = semantic similarity between query and response against ideal answer. Coherence = logical flow and consistency within response.

- Data Sources: User query, model response, ideal answer from evaluation set (for batch) or user thumbs-up/down (for online).
- Thresholds: Relevance ≥ 0.85; Coherence ≥ 0.80 (all apps).

2.5 Content Safety and PII Detection

Definition: Ensures model outputs comply with content safety policies and do not leak PII.

- Computation Method: Azure AI Content Safety API (severity 0–6 scale). PII detection via Microsoft Presidio or Azure AI Language NER. HRHelper: additional attorney-client privilege detection via custom regex + ML classifier.
- Data Sources: Model response, user query (for inbound PII), conversation metadata.
- Thresholds: Hate/Violence/Self-Harm severity ≥ 2 → block. PII leakage > 0 incidents → P0 alert.

3. Infrastructure and Performance Metrics

Metric	Source	AskRetail	StockSense	HRHelper
Request Rate (rpm)	Application Insights	Baseline: 8.3 | Peak: 125	Baseline: 0.3 | Batch: 45	Baseline: 2.2 | Peak: 6.6
Latency p50 (ms)	Application Insights	≤ 800	≤ 8,000 (ad-hoc)	≤ 3,500
Latency p99 (ms)	Application Insights	≤ 2,500	≤ 15,000 (ad-hoc)	≤ 6,000
Error Rate (%)	Application Insights	< 0.1	< 1.0	< 0.5
Apdex Score	Application Insights	≥ 0.94	≥ 0.85	≥ 0.90
CPU Utilization (%)	Container Insights	< 70 (target: 50)	< 80	< 70
Memory Utilization (%)	Container Insights	< 80	< 85	< 80
Pod Restart Rate (/hr)	AKS / Container Insights	< 2	< 5	< 2
API Gateway Throttle Rate (%)	API Management	< 0.5	< 2.0	< 1.0
Cache Hit Rate (%)	API Management / Redis	> 85	> 60	> 75

4. Cost and Quota Metrics

Metric	Source	AskRetail	StockSense	HRHelper

Cost per Conversation ($)	Azure Cost Mgmt + API Mgmt	≤ 0.04	≤ 0.12 (ad-hoc)	≤ 0.08
Cost per 1K Tokens ($)	Azure Cost Mgmt	≤ 0.015	≤ 0.025 (o3-mini)	≤ 0.018
Daily Token Budget (M)	API Management	4.5	1.2	0.8
Monthly Infrastructure ($)	Azure Cost Mgmt	$680K	$420K	$300K
Quota Utilization (%)	API Management	< 85	< 90	< 80
Overage Events (/day)	API Management	< 10	< 50	< 5

5. Continuous Evaluation in Production

Continuous evaluation replays sampled live traffic against candidate models and prompts to detect drift and regression before full rollout.

5.1 Evaluation Runner Configuration

Parameter	AskRetail	StockSense	HRHelper
Sampling Rate	2% of production traffic (min 500/day)	5% of batch jobs (min 50/day)	3% of conversations (min 100/day)
Cadence	Every 4 hours	After each batch run	Every 6 hours
Candidate Comparison	Current vs. candidate model + prompt	Current vs. candidate forecast model	Current vs. candidate policy index + prompt
Metrics Computed	Groundedness, Hallucination, Latency, Relevance, Cost	MAPE, Completion Time, Cost	Groundedness, Escalation Accuracy, Safety, PII
Drift Threshold	≥ 2 metrics degrade > 5%	MAPE increase > 1%	Any safety metric degrades
Action on Drift	Block promotion + alert Platform Ops	Alert CSCO delegate + block model refresh	Block promotion + alert Compliance Officer

5.2 Evaluation Runner Architecture

```
[Production Traffic] --> [Azure Event Hub / Service Bus]
    --> [Sampling Filter (2-5%)]
    --> [Evaluation Orchestrator (Azure Container Instances /
```

```
AKS Job)]
        --> [Replay against Candidate Model (Azure AI Foundry)]
        --> [Azure AI Evaluation SDK Execution]
        --> [Results Storage (Azure Cosmos DB / SQL)]
        --> [Drift Detection Engine]
            -->|Drift Detected| [Alert (Azure Monitor)]
            -->|No Drift| [Baseline Update]
```

6. Dashboard Specifications

6.1 Platform Operations Dashboard (Grafana / Azure Managed Grafana)

- Audience: VP Platform Engineering, SRE on-call, CFO delegate
- Panels: (1) All-apps health status (RAG: Red/Amber/Green), (2) Cross-app latency heatmap, (3) Error rate trend (1h, 24h, 7d), (4) Token consumption rate vs. quota, (5) Cost per hour by app, (6) Active alerts by severity, (7) Deployment status (last 10 releases), (8) AKS cluster resource utilization.
- Refresh: 30 seconds. Time range: Last 4 hours (default), Last 24 hours (shift view).

6.2 Application Owner Dashboard (Azure Monitor Workbooks)

- Audience: CDO delegate, CSCO, CHRO, App Product Managers
- AskRetail Panels: (1) Conversation volume and satisfaction, (2) Groundedness score trend, (3) Hallucination rate, (4) Top failing queries, (5) Revenue attribution (conversions assisted), (6) Latency distribution by region.
- StockSense Panels: (1) Forecast accuracy (MAPE) by category, (2) Batch job completion timeline, (3) Planner engagement (queries per user), (4) Inventory cost impact (carrying cost vs. stockout cost), (5) Model confidence distribution.
- HRHelper Panels: (1) Conversation volume by topic, (2) Escalation rate and accuracy, (3) Content safety incidents (zero tolerance view), (4) Employee satisfaction (CSAT), (5) Audit trail completeness, (6) Data residency compliance status.

6.3 Executive / CFO Dashboard (Power BI)

- Audience: CFO, CDO, Board Technology Committee
- Panels: (1) Monthly AI spend vs. budget (showback), (2) Cost per business outcome (cost per conversion, cost per forecast, cost per HR case resolved), (3) ROI by application (revenue uplift, cost avoidance), (4) Compliance posture (audit findings, open incidents), (5) Deployment velocity (releases per month per app).

7. Alert Severity and Routing Matrix

Severity	Definition	Response Time	Routing	Example Conditions
P0—Critical	Complete service outage or data breach	< 5 min	Page on-call SRE + VP Platform + CISO (if security)	All Tier 1 apps down; PII leak detected; API Gateway 100% error rate
P1—High	Severe degradation or safety violation	< 15 min	Page on-call SRE + App Owner	Latency p99 > 5× SLA for > 10 min; Hallucination > 3%; Content safety severity ≥ 4
P2—Medium	Significant quality or cost anomaly	< 1 hour	Slack/Teams notification + ticket creation	Groundedness drop > 0.05; Cost per request > 150% budget; Semantic drift detected
P3—Low	Minor threshold breach or informational	< 4 hours	Email digest + ticket (next business day)	Cache hit rate < 80%; Token quota > 75%; Single pod restart
P4—Info	Operational data, no action required	N/A	Dashboard only / weekly report	Deployment completed successfully; Baseline evaluation finished; Monthly cost summary

8. Azure Integration Specifications

- Logs: Azure Monitor Logs (Log Analytics Workspace) with Kusto Query Language (KQL). Retention: 90 days hot, 1 year cold (Azure Blob with Archive tier). Structured logging via Serilog (C#) or structlog (Python) in OpenTelemetry format.
- Metrics: Azure Monitor Metrics (platform metrics) + Application Insights Custom Metrics + Prometheus (AKS). Scrape interval: 15s for infrastructure, 60s for business metrics. Retention: 93 days.
- Traces: Application Insights with OpenTelemetry Collector. Sampling: 100% for Tier 1 errors, 10% for success paths (adaptive sampling). Retention: 90 days.

- Alerting: Azure Monitor Alert Rules → Action Groups → PagerDuty/Opsgenie (P0/P1), Microsoft Teams (P2), Email (P3). Smart Detection (auto-anomaly) enabled for AskRetail.
- Dashboards: Azure Managed Grafana for Platform Ops; Azure Monitor Workbooks for App Owners; Power BI for Executive/CFO views.

References: Azure Monitor Documentation 2026, OpenTelemetry Azure Monitor Exporter, Azure AI Evaluation SDK Metrics Guide, Microsoft Well-Architected Framework—Reliability and Performance Efficiency Pillars, NIST AI RMF 1.0 Measurement Appendix.